

## PERFORMANCE PROTOCOL

# PRP OPTIMIZATION

The Double-spin method consistently outperforms single-spin by a significant margin, achieving maximum platelet concentration for superior clinical outcomes.

SINGLE SPIN

**2-3×**

BASELINE

DOUBLE SPIN

**4-6×**

BASELINE

## THE OPTIMIZED PROTOCOL (Amable et al., 2013)

| STEP                                                                          | G-FORCE | TIME   | TEMP | BRAKE |
|-------------------------------------------------------------------------------|---------|--------|------|-------|
| 01 Spin 1 (Soft)                                                              | 300 g   | 5 min  | RT   | Off   |
| → Discard RBC layer, keep plasma + buffy coat                                 |         |        |      |       |
| 02 Spin 2 (Hard)                                                              | 700 g   | 17 min | RT   | Off   |
| → Discard PPP (top), keep platelet pellet, resuspend in ~1 mL residual plasma |         |        |      |       |

## KEY VARIABLES THAT MATTER

### 01. BRAKE OFF

Using the centrifuge brake on the first spin disrupts the buffy coat layer and reduces yield. Always use brake-off or lowest brake setting.

### 02. ROTOR TYPE

Horizontal/swing-bucket rotors give cleaner separation than fixed-angle rotors. Fixed-angle rotors pack platelets against the tube wall.

### 03. TEMPERATURE

Room temperature (20-24°C) is standard. Cold temperatures can activate platelets prematurely.

### 04. TUBE TYPE

Glass tubes activate platelets. Use plastic (polypropylene or PET) tubes with anticoagulant (ACD-A or sodium citrate).

## TECHNICAL SPECIFICATIONS

If your centrifuge only has RPM, use the standard conversion formula:

$$\text{RCF} = 1.118 \times r \times (\text{RPM}/1000)^2$$

For a typical 15 cm radius rotor (r = 150mm):

**300 g**  
~1,340 RPM

**700 g**  
~2,040 RPM

## THE VERDICT

Maximum platelets/mL is achieved with 300 g × 5 min, then 700 g × 17 min, no brake, room temperature, in plastic tubes with ACD-A. Expect ~5-6× the whole blood platelet concentration.

TECHNICAL VERIFICATION REQUIRED  
OPTIMIZED CLINICAL STANDARD

***KADE***

CHIEF PROTOCOL ARCHITECT

PRP OPTIMIZATION PROTOCOL // CREATED BY KADE